Alkali Doped Cryogenic Rare Gas Crystals as a Possible Platform for Solid State Atomic Clocks

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In this project I investigate whether alkali-doped cryogenic rare-gas crystals can serve as a solid-state platform for microwave atomic clocks. Starting from a simple driven two-state system, I derive the projection-noise-limited Allan deviation for Ramsey interrogation and verify the result against the reported stabilities of state-of-the-art microwave and optical clocks. I then show that, using currently demonstrated parameters for Na:Ne doped crystals, one could in principle reach clock stabilities on the order of $\sigma_y(\tau) \sim 10^{-15}/\sqrt{\tau}$, already surpassing primary microwave standards with $\sigma_y(\tau) \sim 10^{-13}/\sqrt{\tau}$. Using Monte Carlo spin-bath simulations of dipole-dipole interactions between dopants, I argue that the observed inhomogeneous coherence times T_2^* in rare-gas matrices are not limited by dopant interaction broadening at the relevant densities, but likely from structural defects in the crystal instead. If these structural effects are improved so that T_2^* approaches the homogeneous T_2 regime, projected stabilities at the level of $\sigma_y(\tau) \sim 10^{-17}/\sqrt{\tau}$ become realistic, and, in an extremely optimistic scenario, stabilities as low as $\sigma_y(\tau) \sim 10^{-20}/\sqrt{\tau}$ are accessible. However, substantial microwave, cryogenic, and crystal-growth engineering challenges make actually reaching these limits extremely difficult in practice.

I. INTRODUCTION

Recent work has demonstrated coherent control over the microwave ground-state hyperfine transitions of sodium atoms embedded in a cryogenic solid neon matrix [1]. This result opens up the possibility I wish to explore in this project: using alkali-doped cryogenic rare-gas crystals as a platform for atomic clocks, specifically microwave atomic clocks. For the alkali systems and hosts I care about here, the optical transitions are strongly shifted and broadened by the matrix environment and, so far, lack the coherent control required to be very useful. This effectively pushes us toward using the ground-state microwave hyperfine transitions instead. Despite that limitation, I will show that, even with the current state of doped cryogenic noble-gas crystals, and with realistic near-term improvements, these systems can already be competitive with existing atomic clocks and have a credible path toward much better performance.

I start with an overview of atomic clocks and derive how they use Ramsey interrogation to lock a local oscillator's frequency to a well-defined atomic transition. In the process, I derive the projection-noise-limited Allan deviation to quantify clock stability and then verify this expression by comparing it to the reported performance of state-of-the-art microwave and optical clocks. This sets a baseline for what type of stability we wish to achieve, while also making clear which parameters transition frequency, coherence time, atom number, and cycle time actually matter and why they matter.

I then give a brief review of doped cryogenic rare-gas crystals, since they are still a fairly niche platform. I explain how their ability to host and simultaneously probe very large numbers of spins makes them attractive for both general precision-measurement experiments and potential clock applications. I focus on the large and recurring discrepancy between the homogeneous coherence time T_2 and the inhomogeneous coherence time T_2^* that is observed in alkali-doped matrices. Using a Monte Carlo–style simulation of a dipole-dipole interactions in a spin bath, I model the inhomogeneous broadening and resulting T_2^* for specific hyperfine transitions at various dopant densities. By comparing these simulated linewidths to experimental data, I argue that, at the dopant densities relevant, the dominant limitation of T_2^* is more likely structural defects in the crystal, rather than fundamental dipole–dipole interactions (though dipole–dipole effects can become a limitation at very high densities).

Finally, I use the measured coherence properties and densities of existing systems to estimate realistic clock stabilities that are achievable right now with a Na:Ne solid-state microwave clock. Using the broadening and coherence arguments developed earlier, I then explore near-term scenarios in which improved crystal growth and structure could push T_2^* closer to the measured T_2, quantifying the corresponding gains in stability. I also examine an extreme fundamental limit case, in which T_2^* approaches T_2 and T_2 approaches $T_1/2$, to show that, in principle, doped-crystal clocks could reach fractional stabilities comparable to or even surpassing those of the best optical lattice clocks.

II. ATOMIC CLOCKS

Atomic clocks are a crucial part of modern science and technology. As the most sensitive clocks available,

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they provide the precise timing required for GPS navigation, telecommunications networks, and the realization of global time standards. The current definition of a second in the SI comes from cesium atomic clocks, specifically defining a second as 9,192,631,770 cycles of the cesium-133 ground-state hyperfine transition [2]. Atomic clocks also enable precision tests of general relativity and searches for physics beyond the Standard Model.

At a basic level, an atomic clock works by repeatedly interrogating a well-defined atomic transition and comparing its frequency to that of a local oscillator. By stabilizing (locking) the local oscillator to the atomic transition, one realizes an extremely stable and accurate frequency reference. For this purpose, one needs an atomic system with two well-defined states separated by a precisely known transition frequency [3].

Most state-of-the-art clocks now rely on optical transitions of atoms trapped in optical lattices. These clocks reach extremely high levels of both accuracy and stability. For example, a strontium optical lattice clock can have a total systematic uncertainty at the level of $\sim 8 \times 10^{-19}$ and a fractional frequency instability on the order of $3 \times 10^{-18}/\sqrt{\tau}$, where τ is the averaging time in seconds [4]. Here, accuracy tells us how close the clock's long-term average frequency is to the true atomic transition, while stability tells us how much the clock's frequency fluctuates in time around that mean. At these levels of uncertainty and stability, one can directly resolve gravitational redshifts over centimeter-scale height changes on Earth [5].

In this project, I will primarily focus on the *stability* of atomic clocks and on the possible stability that a doped cryogenic rare-gas crystal atomic clock could achieve. To do this, I will first set up a simple quantum-mechanical description of an atomic clock as a driven two-state system. Using Ramsey interferometry, one can determine how far the local oscillator is detuned from the resonant transition frequency of the two-state system being probed. I will go through the basic math and expected signal from a standard Ramsey sequence and then show how this leads to an expression for the clock stability in terms of the Allan deviation.

A. Ramsey Interrogation and Allan Deviation

In this section I outline both the standard way one probes an atomic transition with a drive derived from a local oscillator, and how, starting from that interrogation scheme, one can derive the Allan deviation, which quantifies the stability of an atomic clock over an averaging time τ.

In a simple picture, atomic clocks can be idealized as a two-state system with energy splitting $\hbar\omega_0$, whose Hamiltonian is

$$\begin{aligned} H_0 &= \frac{1}{2}\hbar\big(\omega_0 \left|+\right\rangle\left\langle+\right| - \omega_0 \left|-\right\rangle\left\langle-\right|\big) \\ &= \frac{1}{2}\hbar\omega_0\sigma_z = \frac{\hbar}{2}\begin{pmatrix}\omega_0 & 0\\ 0 & -\omega_0\end{pmatrix}. \end{aligned} \tag{1}$$

To actually reference a local oscillator with frequency ω_{LO} to this transition, we use it to drive the transition between these states. Typically, atomic clocks probe either microwave or optical transitions. We can represent this laser or microwave drive under the rotating wave approximation as

$$\begin{aligned} V(t) &= \frac{1}{2}\hbar\Omega_R\big(\left|+\right\rangle\left\langle-\right| e^{-i\omega_{LO}t} + \left|-\right\rangle\left\langle+\right| e^{i\omega_{LO}t}\big) \\ &= \frac{1}{2}\hbar\Omega_R\begin{pmatrix}0 & e^{-i\omega_{LO}t}\\ e^{i\omega_{LO}t} & 0\end{pmatrix}, \end{aligned} \tag{2}$$

where Ω_R is the Rabi frequency.

It is convenient to work in a frame that rotates at the local oscillator frequency. In this rotating frame, the effective Hamiltonian becomes

$$\begin{aligned} \tilde{H} &= \frac{1}{2}\hbar\omega_0\sigma_z - \frac{1}{2}\hbar\omega_{LO}\sigma_z + \frac{1}{2}\hbar\Omega_R\sigma_x \\ &= -\frac{1}{2}\hbar\epsilon\sigma_z + \frac{1}{2}\hbar\Omega_R\sigma_x \\ &= \frac{\hbar}{2}\begin{pmatrix}-\epsilon & \Omega_R\\ \Omega_R & \epsilon\end{pmatrix}, \end{aligned} \tag{3}$$

where we define the detuning

$$\epsilon \equiv \omega_{LO} - \omega_0. \tag{4}$$

Bloch vector description

To characterize how the local oscillator detuning shows up in the measured signal, it is useful to work in the Bloch vector picture. The Bloch vector is the vector of expectation values of the Pauli matrices:

$$\langle\boldsymbol{\sigma}\rangle = \left\langle\psi\right|\boldsymbol{\sigma}\left|\psi\right\rangle. \tag{5}$$

This provides a one-to-one mapping between the state of the two-level system and a point on the unit sphere. In terms of polar and azimuthal angles θ and ϕ, we can write the state as

$$\left|\psi\right\rangle = \cos\left(\frac{\theta}{2}\right)\left|+\right\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)\left|-\right\rangle. \tag{6}$$

To obtain the time evolution of the Bloch vector under the Hamiltonian in Eq. (3), we use the Heisenberg equation of motion:

$$i\hbar\frac{\partial}{\partial t}\langle\boldsymbol{\sigma}\rangle = \langle[\boldsymbol{\sigma}, \tilde{H}]\rangle. \tag{7}$$

Rewriting this gives

$$\frac{\partial}{\partial t}\langle \boldsymbol{\sigma} \rangle = \frac{i}{\hbar}\langle [\tilde{H}, \boldsymbol{\sigma}] \rangle = \frac{i}{2}\langle \big[-\epsilon\sigma_z + \Omega_R \sigma_x, \begin{pmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{pmatrix} \big] \rangle . \tag{8}$$

Using the Pauli commutator relations

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k, \tag{9}$$

we obtain

$$\frac{\partial}{\partial t}\langle \boldsymbol{\sigma} \rangle = \langle \begin{pmatrix} \epsilon\sigma_y \\ -\epsilon\sigma_x - \Omega_R\sigma_z \\ \Omega_R\sigma_y \end{pmatrix} \rangle . \tag{10}$$

This is the Bloch equation for an isolated two-state system with no decay.

To include relaxation, we add phenomenological terms. Longitudinal relaxation with timescale T_1 describes the decay of $\langle\sigma_z\rangle$ to its equilibrium value $\langle\sigma_z\rangle_{eq}$. Transverse relaxation with timescale T_2 describes the decay of $\langle\sigma_x\rangle$ and $\langle\sigma_y\rangle$ to zero. Writing the Bloch vector components as

$$(U, V, W) \equiv \big(\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle \big),$$

the Bloch equations become

$$\dot{U} = \epsilon V - \frac{U}{T_2}, \tag{11}$$

$$\dot{V} = -\epsilon U - \Omega_R W - \frac{V}{T_2}, \tag{12}$$

$$\dot{W} = \Omega_R V - \frac{W - W_{eq}}{T_1}. \tag{13}$$

In an actual clock, we deal with an ensemble of N non-interacting, identically driven atoms. The measured signal corresponds to the ensemble-averaged Bloch vector

$$(U, V, W) = \frac{1}{N}\sum_{j=1}^{N} \big(\langle\sigma_x^{(j)}\rangle, \langle\sigma_y^{(j)}\rangle, \langle\sigma_z^{(j)}\rangle \big). \tag{14}$$

Ramsey Interferometry

We now apply these equations to a Ramsey interferometry sequence. First, prepare all atoms in the state $|-\rangle$, corresponding to

$$(U, V, W) = (0, 0, -1). \tag{15}$$

Next, turn on the drive for a short time $t_p \ll T_1, T_2$, and assume a strong drive $\Omega_R \gg \epsilon$. In this limit we can neglect detuning during the pulse, and the Bloch equations reduce to

$$\dot{U} = 0, \tag{16}$$

$$\dot{V} = -\Omega_R W, \tag{17}$$

$$\dot{W} = \Omega_R V, \tag{18}$$

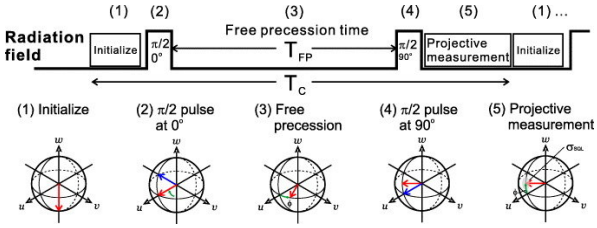


FIG. 1. Bloch-sphere visualization of the Ramsey interrogation sequence used in Sec. II A. Figure taken from [6]

which has the solution

$$U = U_0, \tag{19}$$

$$V = V_0\cos(\Omega_R t_p) - W_0\sin(\Omega_R t_p), \tag{20}$$

$$W = W_0\cos(\Omega_R t_p) + V_0\sin(\Omega_R t_p), \tag{21}$$

for initial values (U_0, V_0, W_0).

Ramsey spectroscopy uses a $\pi/2$ pulse to create an equal superposition of $|+\rangle$ and $|-\rangle$. Starting from Eq. (15) and choosing $\Omega_R t_p = \pi/2$, Eq. (21) gives

$$(U, V, W) = (0, 1, 0). \tag{22}$$

We now turn the drive off and let the system evolve freely for time T_{FP}, during which it accumulates phase due to the detuning ϵ. Physically, this phase accumulation encodes the frequency error of the local oscillator relative to the atomic transition. With $\Omega_R = 0$, the Bloch equations reduce to

$$\dot{U} = \epsilon V - \frac{U}{T_2}, \tag{23}$$

$$\dot{V} = -\epsilon U - \frac{V}{T_2}, \tag{24}$$

$$\dot{W} = -\frac{W - W_{eq}}{T_1}. \tag{25}$$

Assuming $T_{FP} \ll T_1$ (a good approximation for long-lived ground-state coherences) and using the initial condition Eq. (22), one finds

$$(U, V, W) = e^{-T_{FP}/T_2}\big(\sin(\epsilon T_{FP}), \cos(\epsilon T_{FP}), 0\big). \tag{26}$$

To read out this accumulated phase, we apply a second $\pi/2$ pulse with the same drive. Using Eq. (21) again with $\Omega_R t_p = \pi/2$ and the initial values from Eq. (26), we obtain

$$(U, V, W) = e^{-T_{FP}/T_2}\big(\sin(\epsilon T_{FP}), 0, \cos(\epsilon T_{FP})\big). \tag{27}$$

The quantity W, determines the population difference between $|+\rangle$ and $|-\rangle$. The probability to find the system in $|+\rangle$ is

$$P_{|+\rangle} = \frac{1+W}{2} = \frac{1}{2}\big[1 + e^{-T_{FP}/T_2}\cos(\epsilon T_{FP})\big]. \tag{28}$$

To maximize sensitivity to the detuning ϵ, we choose to operate on the side of the Ramsey fringe where the slope is maximized. This corresponds to maximizing

$$\left|\frac{\partial P_{|+\rangle}}{\partial \epsilon}\right|, \tag{29}$$

which occurs when $\epsilon T_{FP} = \pi/2 \pmod{2\pi}$. At that point

$$P_{|+\rangle} = \frac{1}{2}, \quad \left|\frac{\partial P_{|+\rangle}}{\partial \epsilon}\right|_{\max} = \frac{1}{2} e^{-T_{FP}/T_2} T_{FP}. \tag{30}$$

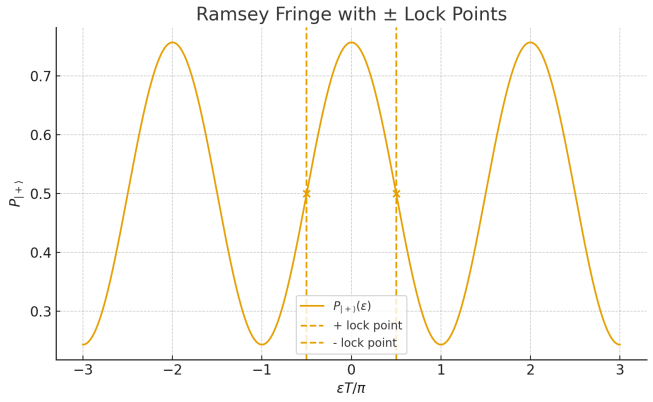


FIG. 2. Ramsey fringe signal $P_{|+\rangle}$ as a function of local-oscillator detuning. The clock is operated on the linear side of the fringe (indicated lock point), where $|\partial P_{|+\rangle}/\partial\epsilon|$ is maximized so that small frequency errors of the local oscillator are converted into the largest possible change in the measured population.

Projection noise and single-shot sensitivity

We now quantify the measurement noise for N independent atoms. For each atom, the probability to be in $|+\rangle$ is $P_{|+\rangle}$, and the outcome is either "in $|+\rangle$" or "not in $|+\rangle$". Thus the number of atoms detected in $|+\rangle$ is distributed as

$$N_{|+\rangle} \sim \text{Binomial}(N, P_{|+\rangle}). \tag{31}$$

The corresponding standard error in the estimated probability $P_{|+\rangle}$ is

$$\sigma_{P_{|+\rangle}} = \sqrt{\frac{P_{|+\rangle}(1 - P_{|+\rangle})}{N}}. \tag{32}$$

At the optimal operating point $P_{|+\rangle} = 1/2$, this becomes

$$\sigma_{P_{|+\rangle}} = \frac{1}{2\sqrt{N}}. \tag{33}$$

We convert this to an uncertainty in the detuning ϵ for a single Ramsey shot using standard error propagation:

$$\sigma_\epsilon^{(1)} = \frac{\sigma_{P_{|+\rangle}}}{\left|\frac{\partial P_{|+\rangle}}{\partial \epsilon}\right|_{\max}} = \frac{1}{e^{-T_{FP}/T_2} T_{FP}\sqrt{N}}. \tag{34}$$

In frequency units, we define the fractional frequency deviation of the local oscillator as

$$y \equiv \frac{\omega_{LO} - \omega_0}{\omega_0} = \frac{\delta\nu}{\nu_0}, \tag{35}$$

where $\nu_0 = \omega_0/2\pi$ is the transition frequency in Hz. For small detunings, the single-shot fractional frequency uncertainty is

$$\begin{aligned} \sigma_y^{(1)} &= \frac{\sigma_\nu^{(1)}}{\nu_0} = \frac{\sigma_\epsilon^{(1)}}{2\pi\nu_0} \\ &= \frac{1}{2\pi\nu_0 e^{-T_{FP}/T_2} T_{FP}\sqrt{N}}. \end{aligned} \tag{36}$$

Allan variance and clock stability

A clock's stability over an averaging time τ is commonly characterized using the Allan variance,

$$\sigma_y^2(\tau) = \frac{1}{2}\left\langle (\overline{y}_{n+1} - \overline{y}_n)^2 \right\rangle, \tag{37}$$

where $\overline{y}_n$ is the average fractional frequency over the nth time interval of duration τ, and $\langle\cdots\rangle$ denotes a statistical (classical) average over many realizations. Here $\langle\cdots\rangle$ is *NOT* a quantum mechanical expectation value.

Each interval of duration τ contains

$$m = \frac{\tau}{T_c} \tag{38}$$

independent Ramsey interrogations, where T_c is the time of one Ramsey measurement: including preparation, interrogation, and readout. We can then write

$$\overline{y}_n = \frac{1}{m}\sum_{k=1}^{m} y_{n,k}, \tag{39}$$

where $y_{n,k}$ is the single-shot fractional frequency estimate from the kth Ramsey sequence in the nth interval.

To connect Eq. (37) with a variance, define

$$\Delta\overline{y}_n \equiv \overline{y}_{n+1} - \overline{y}_n. \tag{40}$$

By the definition of variance,

$$\begin{aligned} \text{Var}(\Delta\overline{y}_n) &= \left\langle \left(\Delta\overline{y}_n\right)^2 \right\rangle - \left\langle \Delta\overline{y}_n \right\rangle^2 \\ &= \left\langle (\overline{y}_{n+1} - \overline{y}_n)^2 \right\rangle - \left\langle \overline{y}_{n+1} - \overline{y}_n \right\rangle^2. \end{aligned} \tag{41}$$

For a well-locked clock we expect, on average, very little drift over one interval, so

$$\left\langle \overline{y}_{n+1} - \overline{y}_n \right\rangle \approx 0, \tag{42}$$

and

$$\text{Var}(\Delta\overline{y}_n) \approx \left\langle (\overline{y}_{n+1} - \overline{y}_n)^2 \right\rangle. \tag{43}$$

Comparing with Eq. (37), we can write the Allan variance as

$$\sigma_y^2(\tau) \approx \frac{1}{2}\,\mathrm{Var}(\overline{y}_{n+1} - \overline{y}_n). \tag{44}$$

Now expand the variance of the difference. Assuming that different $\overline{y}_n$ statistically independent,

$$\begin{aligned}\mathrm{Var}(\overline{y}_{n+1} - \overline{y}_n) &= \mathrm{Var}(\overline{y}_{n+1}) + \mathrm{Var}(\overline{y}_n) \\ &\approx 2\,\mathrm{Var}(\overline{y}_n),\end{aligned} \tag{45}$$

where in the last step we assume the statistics are the similar for all intervals.

Next, write $\overline{y}_n$ explicitly as an average over m independent shots:

$$\begin{aligned}\mathrm{Var}(\overline{y}_n) &= \mathrm{Var}\left(\frac{1}{m}\sum_{k=1}^{m} y_{n,k}\right) \\ &= \frac{1}{m^2}\sum_{k=1}^{m}\mathrm{Var}(y_{n,k}) \\ &= \frac{1}{m}(\sigma_y^{(1)})^2 = \frac{T_c}{\tau}(\sigma_y^{(1)})^2\end{aligned} \tag{46}$$

Putting this back into 45, and then 44, we have the Allan deviation

$$\begin{aligned}\sigma_y(\tau) &= \sqrt{\sigma_y^2(\tau)} \\ &\approx \sqrt{\frac{1}{2}\,\mathrm{Var}(\overline{y}_{n+1} - \overline{y}_n)} \\ &= \sigma_y^{(1)}\sqrt{\frac{T_c}{\tau}} \\ &= \frac{1}{2\pi\nu_0 e^{-T_{FP}/T_2}T_{FP}}\sqrt{\frac{T_c}{N\tau}}.\end{aligned} \tag{47}$$

We want this Allan deviation to be as small as possible, since it sets the smallest resolvable fractional frequency difference over the averaging time τ. The T_{FP}-dependence enters only through the factor $e^{-T_{FP}/T_2}T_{FP}$ in the denominator. To maximize $e^{-T_{FP}/T_2}T_{FP}$, chose

$$T_{FP} = T_2. \tag{48}$$

With this choice, the projection-noise-limited Allan deviation of the atomic clock becomes

$$\sigma_y(\tau) = \frac{1}{2\pi\nu_0 T_2}\sqrt{\frac{T_c}{N\tau}}. \tag{49}$$

Verifying Our Allan Deviation

From the final result in Eq. (49), we can see directly how to improve the stability of an atomic clock:

- increasing the clock transition frequency ν_0,
- increasing the coherence time T_2,
- increasing the number of atoms N interrogated per cycle,
- decreasing the Ramsey cycle time T_c (i.e. reducing dead time).

In the ideal limit $T_c \to T_2$, so the stability is essentially set by three parameters: ν_0, T_2, and N. Different clock architectures push on these levers in different ways.

At a broad level, modern atomic clocks fall into a few main classes (there are other classes, but we won't focus on those in this report).

- Microwave fountain clocks (typically cesium or rubidium) use cold atoms launched in a fountain and interrogate the ground state hyperfine transition [7].
- Vapor-cell clocks. Common in industry as they are cheap and compact.
- Optical clocks push to much higher transition frequencies, either using single trapped ions or many atoms trapped in an optical lattice [4].

In all these classes, the Ramsey interrogation discussed above is typically used, and Eq. (49) should give a reasonable projection-noise-limited stability.

As a microwave example, consider the current primary frequency standard: NIST-F4 [7]. This cesium fountain probes the $\nu_0 \approx 9.192$ GHz ground-state hyperfine transition. The reported Ramsey interrogation time is $T_R \approx 0.5$ s, which we will assume as T_2 in our model, and we take $T_c \approx T_2$. The effective atom number per cycle is on the order of $N \sim 10^4$–10^5, but here we will assume $N = 10^4$. Plugging these values into Eq. (49) gives

$$\sigma_y(\tau) \approx \frac{1}{2\pi\nu_0 T_2}\sqrt{\frac{T_c}{N\tau}} \approx 2.5\times10^{-13}/\sqrt{\tau}, \tag{50}$$

to be compared with the reported stability [7]

$$\sigma_y^{(\text{NIST-F4})} \approx 1.5\times10^{-13}/\sqrt{\tau}. \tag{51}$$

For an optical example, we'll look at a strontium optical lattice clock [4]. This clock interrogates an optical transition at $\nu_0 \approx 4.35\times10^{14}$ Hz, with a Ramsey free-precession time $T_2 \equiv T_R \approx 1$ s, a cycle time $T_c \approx 2.43$ s, and an effective atom number $N \approx 4\times10^4$. Inserting these values into Eq. (49) gives

$$\sigma_y(\tau) \approx \frac{2.85\times10^{-18}}{\sqrt{\tau}}, \tag{52}$$

which matches the reported instability

$$\sigma_y(\tau) \approx \frac{3\times10^{-18}}{\sqrt{\tau}} \tag{53}$$

These two examples show that the simple projection-noise-limited expression in Eq. (49) reproduces the performance of both a state-of-the-art microwave fountain

clock and a state-of-the-art optical lattice clock. In the rest of this work, I will use the same expression to estimate the potential stability of doped cryogenic rare-gas crystal microwave clocks, first using currently reported coherence times, and then exploring how the stability would change if T_2^* could be be improved, or if we grow larger crystals, boosting N.

Verifying our Allan Deviation

From the final results of the derivation above, we can begin to gauge how one improves the stability of an atomic clock by controlling a few primary parameters:

- Increasing clock transition frequency ν_0
- Increasing spin-spin relaxation coherence time T_2
- Increase number of atoms interrogated N
- Decreasing dead time and resulting Ramsey cycle time T_c

The best you can do for T_c is have it approach T_2, however, so we are really limited by three variables: ν_0, T_2, and N.

In state-of-the-art microwave clocks, such as the current primary frequency standard, NIST-F4 [7], report a frequency stability of

$$\sigma_y^{(\text{NIST-F4})} = 1.5 \times 10^{-13}/\sqrt{\tau}. \tag{54}$$

To confirm the validity of our derived stability equation 49, we will plug in the values reported for NIST-F4:

$$\begin{aligned} &\nu_0 \approx 9.192\text{GHz} \\ &T_2 \equiv T_R = 0.5\text{s} \\ &N \lesssim 10^5 \rightarrow N \equiv 10^4 \end{aligned}$$

therefore, using 49

$$\sigma_y(\tau) = 2.5 \times 10^{-13}/\sqrt{\tau}. \tag{55}$$

Similarly, the state-of-the-art optical clocks can report stability of [4]

$$\sigma_y(\tau) = 3 \times 10^{-18}/\sqrt{\tau}, \tag{56}$$

while using

$$\begin{aligned} &\nu_0 \approx 4.35 \times 10^{14}\text{Hz} \\ &T_2 \equiv T_R \approx 1\text{s} \\ &T_c = 2.43\text{s} \\ &N = 4 \times 10^4. \end{aligned}$$

Plugging these values into 49 we get

$$\sigma_y(\tau) = 2.85 \times 10^{-18}/\sqrt{\tau}. \tag{57}$$

We now have both confirmed the validity of our derived Allan Deviation for clock stability, as well as set some benchmarks for state-of-the-art microwave and optical atomic clocks that we can compare to.

III. DOPED CRYOGENIC RARE GAS CRYSTALS

Doped inert cryogenic crystals offer a promising platform for fundamental physics research, including high-precision measurements and searches for physics beyond the Standard Model. By encasing atoms or molecules (dopants) inside an inert cryogenic crystal, one creates an "artificial vacuum." (Fig. 3). This pristine environment enables precise measurements of quantum properties such as the electron electric dipole moment (eEDM), as well as clock-like transitions suitable for precision frequency metrology [8, 9]. Electron EDM measurements are among the most sensitive probes for new physics that could explain some of the biggest unanswered questions in fundamental physics, such as the matter–antimatter asymmetry. Additionally, these doped crystals have potential applications in dark matter searches and in quantum information science for quantum sensors and simulators [1, 9]. Recently, coherent control of the microwave hyperfine transition in Na atoms embedded in solid neon has been demonstrated, indicating that these crystals can support the essential ingredients of a microwave atomic clock [10]. In this work we therefore focus on doped cryogenic rare-gas crystals as a solid-state platform that can simultaneously host large ensembles of alkali atoms with long coherence times, making them attractive candidates for microwave atomic clocks.

A major source of interest in doped cryogenic crystals is their ability to greatly increase statistical sensitivity, since they can isolate a large number of atoms or molecules for simultaneous measurements. For example, one of the most competitive eEDM measurements, ACME-II, spent $\sim$ 350 hours collecting data in a beam of a few hundred thousand molecules flowing through every second [11]. Doped crystals can contain orders of magnitude more molecules: a doped cryogenic crystal with a dopant density of $10^{16}\,\text{dopants/cm}^3$ could, in principle, produce an eEDM measurement with the same uncertainty as ACME-II in seconds instead of weeks. The same argument applies directly to atomic clocks. The projection-noise-limited Allan deviation derived in Sec. II A improves as $1/\sqrt{N}$ with the number of atoms N, so the large ensembles naturally accessible in doped crystals are an enormous asset for clock sensitivity.

To exploit this advantage, however, the dopants must maintain coherence for a sufficiently long ensemble transverse spin relaxation time T_2^*. For a dopant density of $10^{16}\,\text{dopants/cm}^3$, achieving $T_2^* \gtrsim$ ms should be enough to match the statistical sensitivity of leading beam experiments. Previous work has achieved $T_2^* \sim 1$ ms with cesium-doped helium crystals, but the pressure required to solidify helium makes growth of crystals with high dopant density technically intractable [12]. Other groups have grown neon, parahydrogen, and argon crystals doped with alkali atoms at the desired dopant densities but with insufficient T_2^* times, typically $T_2^* \sim$ 10–100 μs [13–16]. In the context of clocks, these short

T_2^* times directly cap the Ramsey free-precession time T_{FP}, and therefore limit the achievable Allan deviation through Eq. (49).

Despite their potential, doped crystals currently suffer from what many believe are structural defects and inhomogeneities that limit their performance [9, 14, 16]. This is, however, somewhat debated, and will be discussed later. While much research has gone into exploring the use of different dopants and host crystals and the resulting measurements that can be made, less has been done to fully understand the relationship between crystal growth parameters and the resulting crystal structure and coherence properties. Dopants that experience different local geometries and local fields within the same crystal dephase relative to one another, decreasing the size of the collective signal and reducing T_2^* even when the single-site homogeneous T_2 is long. Hosting dopants inside a nearly perfect single crystal, or at least strongly suppressing structural disorder, should provide a more uniform local environment and thereby increase the ensemble coherence time.

Growth parameters such as beam velocity, dopant density, growth rate, and growth-substrate temperature all interact and affect the crystal. Beam velocity influences how dopants incorporate into the lattice structure, and the surface dynamics of incorporation are only qualitatively understood. Most groups have chosen to grow crystals at rates of order 0.5–2 μm/min without much systematic justification [15, 16], and optical clarity has often been an issue, hindering the use of optical techniques [9, 15]. Although poor optical quality is typically attributed to growth-substrate temperature, it has not been systematically studied, and it is likely that multiple growth parameters contribute. In general, a systematic study of the relationship between growth conditions and crystal properties is still missing. For microwave atomic clocks based on alkali dopants, understanding and controlling this growth–structure–coherence relationship is important. and needed to engineer crystals that simultaneously support high dopant densities, long T_2^*, and narrow clock transitions suitable for the Ramsey interrogation schemes analyzed in Sec. II A.

A. Homogeneous vs inhomogeneous coherence: T_1, T_2, and T_2^*

Within doped cryogenic rare-gas crystals we see a huge disparity between the T_1, T_2, and T_2^* coherences. These disparities are important to understand if one wants to make these crystals into a useful platform. In this section I will start by explaining the difference between these coherence times and the coherences we actually see in these crystals. I will then point out the large gap between T_2 and T_2^*, and discuss the ongoing debate about whether this inhomogeneous dephasing is primarily caused by dipole–dipole interactions of the dopants, or by structural defects and disorder in the crystal.

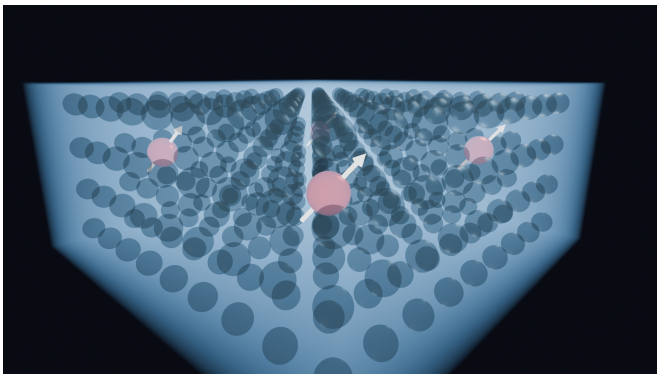


FIG. 3. Cartoon of an alkali-doped rare-gas crystal. The blue array represents the host rare-gas lattice, while the red spheres with arrows represent embedded dopant atoms and their spin.

In the Bloch-vector picture, T_1 is the longitudinal relaxation time. It sets how quickly the population difference $W = \langle\sigma_z\rangle$ relaxes back to its thermal equilibrium value W_{eq}, i.e. the $|+\rangle$ or $|-\rangle$ state. Physically, T_1 is set by spin-lattice relaxation, or interactions between the dopants and phonons in the host crystal. T_2 is the homogeneous transverse relaxation time. It characterizes the decay of the transverse components $U = \langle\sigma_x\rangle$ and $V = \langle\sigma_y\rangle$ and the loss of spin phase. The exact cause of T_2 in crystals does not seem to have a clear explanation yet. T_2 is also what you extract when doing Hahn echo to refocus our ensemble of spins.

In a real experiment we never probe a single dopant, but an ensemble of N dopants that each see slightly different local environments. Variations in these local environments, possibly from crystal defects or dopant-dopant interactions, shift the resonant frequency of each dopant by a slightly different amount. Even if each individual spin has a long homogeneous T_2, these static frequency offsets cause the ensemble-averaged transverse magnetization to dephase much faster. This ensemble dephasing time is what we call T_2^* and is what we are limited by when doing Ramsey interferometry, at least in the case of doped crystals.

$$\frac{1}{T_2^*} \approx \frac{1}{T_2} + \frac{1}{T_{\text{inh}}}, \tag{58}$$

where T_{inh} is an effective time set by the width of the static frequency distribution. In most doped rare-gas crystals to date, $T_{\text{inh}} \ll T_2$, so T_2^* is dominated by inhomogeneous broadening rather than by the true homogeneous T_2. However, in principle, if one can grow crystals that significantly limit inhomogeneous effects causing this different dephasing between environmental cites, one could achieve

$$T_2^* \to T_2. \tag{59}$$

This would allow us to reach the ms coherence times mentioned earlier. In an ideal limit, one is fundamentally

limited to

$$T_2^* \to T_2 \to \frac{1}{2}T_1. \tag{60}$$

However, we are a long way from realizing this limitation in these crystals.

B. Modeling inhomogeneous broadening and T_2^*

One ongoing question in the doped cryogenic rare-gas crystal community is what primarily limits the ensemble coherence time T_2^*. Short T_2^* has been attributed to interactions with the host matrix, static inhomogeneity in the crystal structure (strain, defects, multiple trapping sites), or magnetic dipole-dipole interactions between dopants. Different groups also operate at different dopant densities and with different species, which may result in some disagreement as will become evident from work below.

To get a handle on at least one of these possible mechanisms, I built a Monte Carlo simulation that isolates dipole-dipole interactions between neighboring dopants to see if dipole-dipole interactions were the only source of inhomogeneous broadening, what T_2^* would we expect at a given dopant density and a given hyperfine transition? By comparing this dipolar-only prediction to experimental linewidths/T_2^* values, we can see if dipole-dipole interactions are able to explain the observed broadening or if/when an additional mechanism must be included.

A full description of the model, assumptions, and implementation details is given in Appendix A. Here I only summarize the essential concepts and how they connect to the figures in this section.

1. *Monte Carlo dipole-dipole broadening model: overview*

Because I am interested in doped crystals as a platform for microwave atomic clocks, I focus on transitions within the hyperfine ground-state manifold of alkali atoms. In practice, this means choosing a two state transition such as

$$|F_\mathrm{g}, m_{F,\mathrm{g}}\rangle \to |F_\mathrm{e}, m_{F,\mathrm{e}}\rangle\,,$$

computing the corresponding effective Zeeman coefficient for that transition, and then asking how a random bath of neighboring magnetic dipoles shifts that transition frequency.

The simulation proceeds in three main steps:

1. **Sample a spatial configuration of dopants.** For a chosen dopant density ρ (in atoms/cm^3) and cubic box size L, I draw a Poisson-distributed number of dopants with mean $\bar{N} = \rho L^3$ and place them uniformly at random in the cube. One "central" dopant at the origin is designated as the test atom and all others act as its magnetic environment. An example realization of such a configuration is shown in Fig. 4.

2. **Assign spin states and magnetic moments.** For each neighboring dopant I assign a hyperfine manifold F and Zeeman sublevel m_F using simple "pumping efficiency" parameters that mimic optical pumping into a desired $F_\mathrm{pump}, m_{F,\mathrm{pump}}$ state. I also allow for imperfect alignment of the magnetic moments to the quantization axis via an "alignment efficiency" parameter. In practice, the results turn out to be relatively insensitive to the exact pumping efficiencies and rely almost entirely on dopant densities and the two state transition we probe.

3. **Compute the dipole magnetic field and frequency shift.** Each dopant is treated as a point magnetic dipole. The magnetic field from each dipole at the origin is computed using the standard dipole magnetic field formula, and the z-component is summed over all neighbors to obtain B_z at the central site. This B_z is then converted into a frequency shift $\Delta\nu$ of the chosen transition from a first-order Zeeman shift.

Repeating this procedure over tens to hundreds of thousands of random configurations generates a distribution of frequency shifts. A typical histogram of these shifts is shown in Fig. 5. In the analysis, this histogram is fit to a Lorentzian to extract an effective full width at half maximum (FWHM) Γ, which I then convert to an inhomogeneous coherence time via

$$T_2^* = \frac{1}{\pi\Gamma}.$$

This T_2^* is interpreted as the ensemble coherence time for the chosen transition at the chosen density, in the absence of any other inhomogeneous broadening mechanism.

2. *Density scaling from the dipolar model*

Once the basic Monte Carlo model is in place, it is straightforward to repeat the calculation at multiple dopant densities and look at how the predicted T_2^* (or linewidth) scales with ρ. In the model, the relevant scale for the local magnetic field is set by the typical distance to the nearest neighbors, which shrinks as ρ increases. As a result, the simulated inhomogeneous linewidth grows approximately linearly with density, and the corresponding T_2^* decreases roughly as $1/\rho$.

Figure 6 shows said trend extracted from the simulation. At low densities the magnetic dipole fields are small and the predicted linewidths are narrower, at high densities magnetic dipole fields are relatively large resulting in broader predicted linewidths. If real crystals were dipole-dipole limited, we would expect a strong and fairly predictable dependence of linewidth on dopant density; however, we don't fully see this, which I will continue to show below.

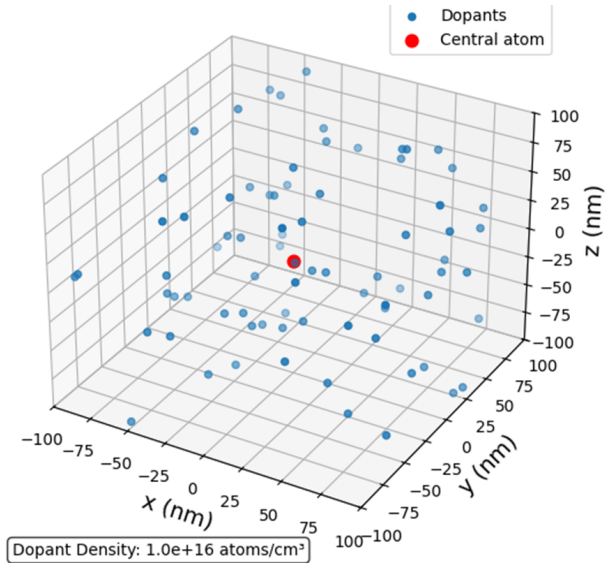


FIG. 4. Example Monte Carlo realization of a dopant configuration used in the dipole-dipole inhomogeneous broadening model. The central test dopant is fixed at the origin, while all other dopants are placed randomly uniformly in a cubic volume of side L at the target density ρ. For each realization, the dipole magnetic field from all neighbors is summed at the origin and converted to a hyperfine transition frequency shift of the central test dopant.

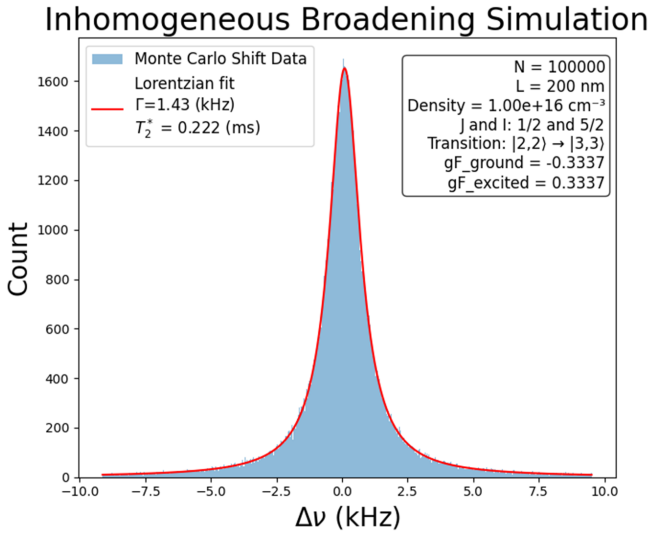


FIG. 5. Example Histogram $H(\Delta\nu)$ from counts of Monte Carlo frequency shifts. Each bin counts how often a given shift $\Delta\nu$ appears over many random configurations of neighboring dopants. This histogram is fit to a Lorentzian to extract an effective inhomogeneous linewidth Γ, which is then mapped to a coherence time $T_2^* = 1/(\pi\Gamma)$.

3. *Inhomogeneous broadening model vs experimental data*

To see where the dipolar model works and where it fails, I compare its predictions to several experimental systems where both the dopant density and an inhomogeneous linewidth or T_2^* have been reported. These comparisons naturally divide into two regimes:

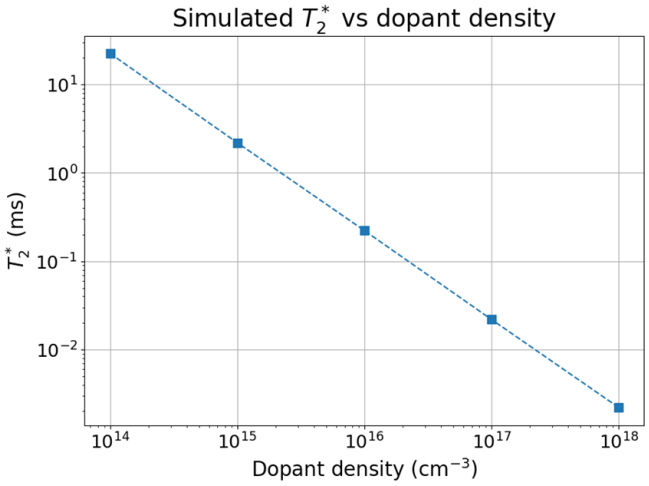


FIG. 6. Simulated dependence of the coherence time T_2^* on dopant density. This shows up as an approximately linear increase of inhomogeneous linewidth with density and an approximate $T_2^* \propto 1/\rho$ scaling. Experimental data, however, often show a much weaker dependence on density, especially in the low-density regime.

- a "high-density" regime, where dopant densities are $\rho \gtrsim 10^{17}$ atoms/cm^3
- a "low-density" regime, with $\rho \sim 10^{16}$ atoms/cm^3

The high-density regime turns out to be reasonably well described by the dipole-dipole interaction broadening model, at least to within an order of magnitude. At low densities, however, the model fails, predicting linewidths at least two orders of magnitude lower than what is observed. This suggests that in most existing doped rare-gas crystals, dipole-dipole interactions are *not* the main source of inhomogeneous broadening.

High dopant densities: solid parahydrogen As a high-density test case, I use the data from Upadhyay *et al.* [14], who study several alkali species embedded in solid parahydrogen. In that work they investigate ^{39}K, ^{85}Rb, ^{87}Rb, and ^{133}Cs, and for each species they probe the same hyperfine transition

$$|F = 2, m_F = -1\rangle \rightarrow |F = 2, m_F = 0\rangle \,.$$

They report that changing the dopant density does not significantly affect the linewidth until a certain upper density limit is reached, beyond which the broadening becomes much worse. I take these "limit" densities and the corresponding linewidths as inputs for testing the Monte Carlo model.

The approximate dopant densities used for each species are summarized in Table I.

Using these densities and the reported transition, I run the Monte Carlo model for each species, extract a

Species	Dopant density ρ (atoms/cm^3)
^{39}K	5×10^{16}
^{85}Rb	$\sim 10^{18}$
^{87}Rb	$\sim 10^{18}$
^{133}Cs	10^{17}

TABLE I. Approximate dopant density concentrations used in the solid parahydrogen experiments of Ref. [14].

linewidth, and compare it to their experimental values. The results are shown in Fig. 7. The key point is that, for all four species, the model and experimental data agree to within roughly an order of magnitude. Given the simplicity of the model and the various uncertainties in the exact state distribution and local environment, this level of agreement is fairly good.

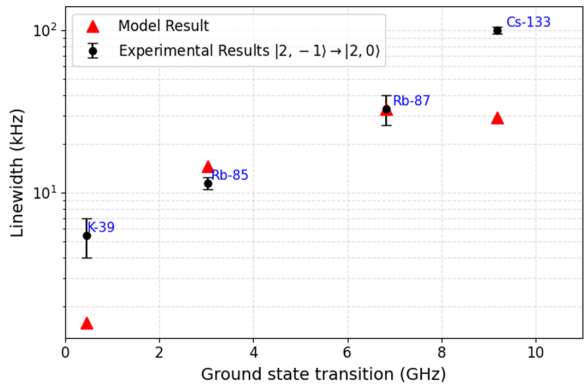


FIG. 7. Comparison between experimentally reported inhomogeneous broadening (or equivalently T_2^*) and the dipolar-limited values predicted by the Monte Carlo model for ^{39}K, ^{85}Rb, ^{87}Rb, and ^{133}Cs in solid parahydrogen at the high dopant densities listed in Table I. For each species the model uses the same $|F = 2, m_F = -1\rangle \rightarrow |F = 2, m_F = 0\rangle$ hyperfine transition studied in Ref. [14]. The agreement is within an order of magnitude across all species, indicating that at these high densities dipole-dipole interactions can plausibly be a leading contributor to the observed inhomogeneous broadening.

Upadhyay *et al.* also report linewidths for several neighboring m_F transitions within the $F = 3$ ground-state manifold of ^{85}Rb in solid parahydrogen [14]. Running the same Monte Carlo procedure for each of these transitions at their quoted densities, I again find that the model reproduces the overall scale and relative ordering of linewidths to within an order of magnitude, as shown in Fig. 8. This further supports the idea that, at very high dopant densities, dipole-dipole interactions could become an important or dominant source of inhomogeneous broadening.

Low dopant densities: Rb:Ne and Na:Ne The situation looks very different when we move to the lower dopant densities. Two important examples here are Rb in solid neon and Na in solid neon.

Dargyte *et al.* study ^{85}Rb in solid neon at densities

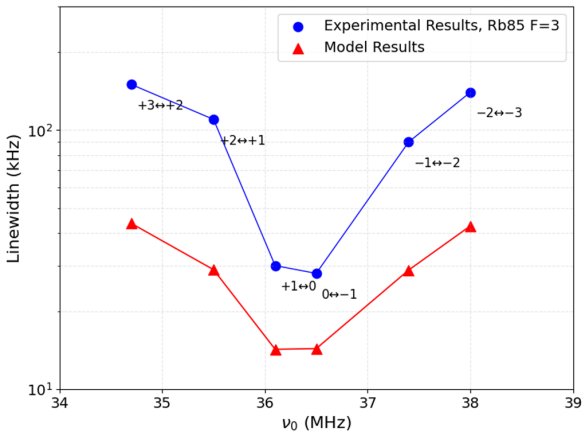


FIG. 8. Simulated versus experimental inhomogeneous broadening for several neighboring m_F transitions within the $F = 3$ ground-state manifold of ^{85}Rb in solid parahydrogen, using values reported in Ref. [14].

around $\rho \sim 10^{16}$ atoms/cm^3 and report linewidths for several neighboring m_F transitions within the $F = 3$ ground-state manifold similar to above [16]. Using their densities and transitions as inputs to the Monte Carlo model, I obtain dipole-dipole broadened linewidths that are almost two orders of magnitude smaller than the experimental values. The comparison is summarized in Fig. 9.

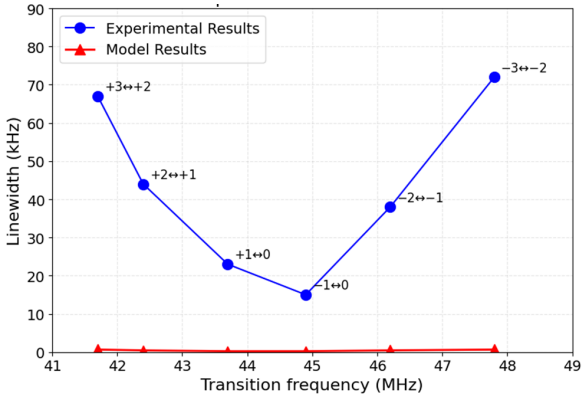


FIG. 9. Comparison between experimental inhomogeneous linewidths for ^{85}Rb in solid neon at densities $\rho \sim 10^{16}$ atoms/cm^3 [16] and the dipole-dipole broadening limited values predicted by the Monte Carlo model. At these "low" densities, the simulated broadening is nearly two orders of magnitude smaller than observed.

This discrepancy is even more striking when we notice that the Rb:Ne linewidths at $\rho \sim 10^{16}$ atoms/cm^3 are comparable to the Rb:parahydrogen linewidths at densities that are roughly two orders of magnitude higher. If dipole-dipole interactions were the dominant broadening mechanism in both systems, we would expect a strong density dependence across these experiments, which is not observed.

An even more relevant example for this project is Kanagin's Na:Ne microwave transition, as we want to access at least microwave transitions to make an atomic clock. In this work, Kanagin grows Na-doped neon crystals at densities of roughly $\rho \sim 3 \times 10^{16}\,\mathrm{atoms/cm}^3$ and probes the hyperfine transition

$$|F = 1, m_F = 1\rangle \rightarrow |F = 2, m_F = 2\rangle$$

in ^{23}Na [10]. The reported inhomogeneous linewidth for this transition is about 989 kHz. Running my Monte Carlo dipolar model at the same density and for the same transition, I obtain a simulated linewidth of only 2.29 kHz. The comparison is shown in Fig. 10.

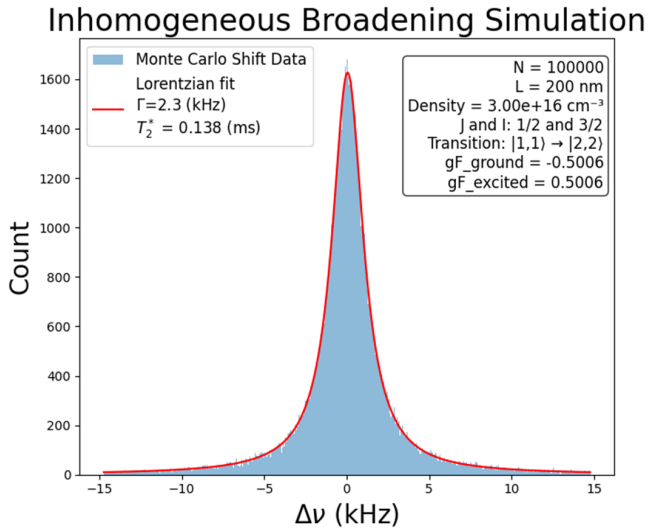


FIG. 10. Dipole-dipole inhomogeneous broadening model for the $|F = 1, m_F = 1\rangle \rightarrow |F = 2, m_F = 2\rangle$ microwave transition of ^{23}Na in solid neon at a dopant density of $\rho \sim 3 \times 10^{16}\,\mathrm{atoms/cm}^3$ [10]. The experimental inhomogeneous linewidth is approximately 989 kHz, whereas the model predicts only about 2.29 kHz, more than two orders of magnitude lower.

Implications for the dominant broadening mechanism and clock transitions Taken together, these comparisons paint a consistent picture:

- At very high dopant densities, the dipole-dipole broadening model predicts linewidths that are comparable (within an order of magnitude) to experiment. In this regime, it is plausible that dipole-dipole interactions are a main sources of inhomogeneous broadening.

- At lower dopant densities, the same model predicts linewidths that are much smaller than observered, often by factors of 100 or more. Here, something other than dipole-dipole interactions between dopants must be dominating the inhomogeneous broadening.

The near insensitivity of the measured linewidths to dopant density across different hosts and densities, combined with the large discrepancy between simulated and observed linewidths at low ρ, strongly supports the idea that structural disorder and multiple trapping sites in the crystal are the main culprits for the short T_2^* observed.

This conclusion matters for doped-crystal atomic clocks for two main reasons. First, if dipole-dipole interactions were the dominant broadening mechanism, then probing a clock transition with $m_F = 0$ on both states (which is first-order insensitive to magnetic fields) would be expected to substantially improve T_2^* compared to neighboring Zeeman-sensitive transitions. However, we see that this likely will not be the case. Second, if the broadening is structural in origin, then we should be able to improve it with better growth techniques. Allowing vast improvements in our coherence including for these clock transitions. By understanding and controlling the growth conditions, trapping sites, and strain in the host crystal, it should be possible to systematically reduce this structural inhomogeneity and push T_2^* much closer to T_2.

This is exactly the direction being pursued in the Odom lab at Northwestern. The Monte Carlo results here suggest that if crystal defects and site disorder can be significantly reduced, then there is no obvious fundamental dipole-dipole interaction broadening preventing T_2^* from reaching the millisecond regime, given our dopant densities are low enough as shown in Fig. 6.

IV. ATOMIC CLOCK STABILITY IN DOPED CRYSTALS

Now that we have covered the fundamentals of atomic clock stability, the structure and limitations of doped cryogenic rare-gas crystals, and the origin of possible decoherence in these systems, we can finally estimate what clock performance is realistically achievable today, and what could be possible if crystal quality improves.

We start with a sodium-doped neon (Na:Ne) crystal. At present, this is the only demonstrated platform (I am aware of) in which the ground-state microwave hyperfine transition of an alkali atom that has been coherently accessed inside a rare-gas solid [1]. Kanagin *et al.* grow Na:Ne crystals at a dopant density of roughly $\rho \approx 3 \times 10^{16}\ \mathrm{cm}^{-3}$ and report an inhomogeneous linewidth of $\Gamma = 989$ kHz for the microwave hyperfine transition. Using the usual Lorentzian conversion $T_2^* = 1/(\pi\Gamma)$ gives an inhomogeneous coherence time of $T_2^* \approx 0.3\ \mu$s. Independent Hahn-echo measurements of the same crystal give a homogeneous $T_{2,\mathrm{Hahn}} = 0.93$ ms, and inversion-recovery measurements give a spin–lattice relaxation time of $T_1 \approx 10$ min [10]. Finally, the free-space ground-state hyperfine splitting of ^{23}Na is $\nu_0 \approx 1.772$ GHz [17], which we will use for stability estimates.

We summarize the relevant values in Table II.

We now insert these values into the projection-noise-limited Allan deviation derived in Eq. 49. Because the

TABLE II. Na:Ne crystal parameters used for clock stability estimates

ρ (cm^{-3})	T_2^* (μs)	$T_{2,\mathrm{Hahn}}$ (ms)	T_1 (min)	ν_0 (GHz)
3×10^{16}	~ 0.32	0.93	10	1.772

system is inhomogeneously limited, the Ramsey free-precession time, T_{FP}, cannot exceed T_2^*, and we assume the cycle time approaches $T_c \to T_2^*$. We also assume a crystal with a volume of ~ 1 cm^3 for simplicity. However, the Odom lab has achieved cm scale, optically clear, pure Argon crystals, making this volume a somewhat reasonable possibility, Fig. 11.



FIG. 11. cm scale optically clear Argon crystal grown by Daniel Lutes in the Odom lab in Northwestern's Center for Fundamental Physics. Demonstrating the achievability of cm scale doped crystals

so at the reported density we have

$$N = \rho V = 3\times10^{16}$$

atoms. Using Eq. 49, the Allan deviation simplifies under our assumptions to

$$\sigma_y(\tau) = \frac{1}{2\pi\nu_0}\frac{1}{\sqrt{T_2^* N}}\frac{1}{\sqrt{\tau}}.$$

Plugging in the current Na values gives

$$\sigma_y(\tau) \approx 10^{-15}/\sqrt{\tau}.$$

Even with the extremely poor present-day inhomogeneous coherence time ($T_2^* \sim 0.3$ μs), the achievable clock stability already surpasses the current state-of-the-art microwave primary standard, NIST-F4, which reports

$$\sigma_y^{(\text{NIST-F4})}(\tau) = 1.5\times10^{-13}/\sqrt{\tau}$$

[7]. The sheer number of atoms that can be contained and interrogated simultaneously in these crystals is astonishing, allowing them to become competitive even in its current, suboptimal form. The dense ensemble compensates for its terrible coherence time.

We now consider a scenario where crystal inhomogeneity is reduced enough that the ensemble coherence approaches the homogeneous limit, i.e. $T_2^* \to T_{2,\mathrm{Hahn}} \approx 1$ ms. Repeating the estimate gives

$$\sigma_y(\tau) \approx 10^{-17}/\sqrt{\tau},$$

which pushes the system into the stability regime occupied by optical lattice clocks [4].

Finally, if we consider the fundamental limit where the crystal reaches the spin–lattice limit

$$T_2^* \to T_2 \to \tfrac{1}{2}T_1,$$

with $T_1 \approx 10$ min, the system would support a Ramsey time of order few minutes. Inserting this value yields

$$\sigma_y(\tau) \approx 3\times10^{-20}/\sqrt{\tau}.$$

This represents an extreme, but physically allowed, limit of the doped crystal atomic clock platform. A stability surpassing even the best optical clocks currently in operation. Achieving such a regime is obviously not a realistic near-term goal, but it does show how promising these doped crystals could be in the long term with their incredible fundamental limits.

Even now, however, there are straightforward paths for improvement. Sodium has a relatively small hyperfine splitting ($\nu_0 \approx 1.77$ GHz), which directly limits fractional stability. If coherent control can be demonstrated in something like a cesium-doped crystals, with the larger cesium hyperfine frequency ($\nu_0 \approx 9.19$ GHz), we would immediately improve stability by roughly a factor of five. Increasing the crystal volume or raising the dopant density also provides another direct lever, as

$$\sigma_y(\tau) \propto 1/\sqrt{N}.$$

Overall, even the conservative estimates already demonstrate that doped rare-gas crystals can, in principle, rival or exceed microwave clock performance simply by exploiting the huge ensemble size. With improvements to crystal uniformity, these systems could begin to approach the stability regime of optical clocks, and with more aggressive improvements they could theoretically surpass them.

A. Current realistic limitations

While the stability estimates above are exciting, we are still very far from a practical experimental setup that actually achieve these bounds. The Na:Ne set up demonstrated by Kanagin is already a highly nontrivial system, and it still falls short of the assumptions that went into my simple Allan deviation estimates.

First, for state preparation to get the Na ensemble into a reasonably well-defined initial spin state, Kanagin had to cool the system down to abase temperature of about

50 mK [10]. At 3 K, the sodium hyperfine levels are still almost equally populated (roughly 50/50 between $|F = 1\rangle$ and $|F = 2\rangle$) so the ensemble is essentially unpolarized [10]. In principle, some optical pumping scheme could let us get away with higher temperatures, but this has yet to be demonstrated in a rare-gas matrix while simultaneously doing coherent microwave control.

Second, the effective number of spins interrogated in the microwave resonance is nowhere near the naive picture of being able to interrogate the entire crystal. In Kanagin's work, the microwave resonator only overlaps a small mode volume, and only a fraction of the dopants are both thermally polarized and pumped into the desired $|F, m_F\rangle$ state. Kanagin finds that only $N \approx 6.8 \times 10^5$ spins are actually being interrogated by the resonator [10]. That is roughly *ten orders of magnitude* smaller than the $N \sim 10^{16}$ I used in my idealized 1 cm^3 estimate. The resulting stability we can expect then, while also using the current T_2^* times, becomes

$$\sigma_y(\tau) \approx 2 \times 10^{-10}/\sqrt{\tau}.$$

Which, while this stability is still on par with some chip scale clocks, requires much bulkier vacuum and cryogenic chambers, making it not as practical or useful.

Additionally, the microwave engineering problem is nontrivial. Even if we imagine a future crystal with much better structural quality and suppressed inhomogeneity, we still need a microwave cavity or waveguide structure that can couple to a large fraction of the dopants, provide a homogeneous microwave field over that cm scale volume, and does not introduce additional technical noise or heating that kills coherence. Kanagin's uses a superconducting resonator with a small mode volume, which is great for achieving strong coupling to spins, but it does work well for larger mode volumes, or at higher temperatures [10].

Overall, the stability numbers I present are not crazy as a fundamental limit, but realistically achieving them is incredibly non-trivial and requires a significant amount of new engineering. It does seem possible, however, that a number of these task could be achieved over the course of a PhD. But until these engineering issues are solved, these doped cryogenic rare-gas crystal atomic clocks remain far off.

V. CONCLUSION

Through this project I have argued that alkali doped cryogenic rare-gas crystals provide a promising platform for solid-state microwave atomic clocks. Through deriving the projection noise limited Allan deviation for clock stability, I have shown that doped crystals, by utilizing their enormous atom number, can achieve stabilities on the order of $10^{-15}/\sqrt{\tau}$, already surpassing the microwave state-of art NIST-F4. And in the extreme fundamental limit could, in theory, achieve order $10^{-20}/\sqrt{\tau}$ stability, surpassing even current state-of-the-art optical clocks. Monte Carlo spin bath simulations have also shown how the current limitation on doped crystals, inhomogeneous broadening, can likely be attributed to structural defects and improved to reach these higher fundamental limits presented. However, there are currently significant engineering issues required to even come close to the "realistic" and lower achievable stabilities I have presented.

Appendix A: Monte Carlo model for dipole-dipole interaction induced inhomogeneous broadening

In this appendix I summarize the Monte Carlo model used to estimate the inhomogeneous broadening from dipole-dipole interactions and the corresponding ensemble coherence time T_2^* for a given dopant density.

1. Transition and Zeeman coefficient

We consider a single atom at the origin and a chosen hyperfine transition

$$|F_\mathrm{g}, m_{F,\mathrm{g}}\rangle \to |F_\mathrm{e}, m_{F,\mathrm{e}}\rangle\,, \tag{A1}$$

within a manifold defined by the electronic angular momentum J and nuclear spin I.

For each hyperfine level F we use the standard hyperfine Landé g-factor, dropping the nuclear term,

$$g_F = g_J \frac{F(F+1) - I(I+1) + J(J+1)}{2F(F+1)}, \tag{A2}$$

with $g_J \simeq 2.0023$ for the alkali ground state, $J = 1/2$. The effective Zeeman coefficient for the chosen transition is taken as

$$\gamma_\mathrm{eff} = \frac{\mu_B}{h}\big(g_{F_\mathrm{e}} m_{F,\mathrm{e}} - g_{F_\mathrm{g}} m_{F,\mathrm{g}}\big), \tag{A3}$$

so a magnetic field B_z at the central atom produces a frequency shift

$$\Delta\nu = \gamma_\mathrm{eff} B_z. \tag{A4}$$

2. Sampling dopant configurations

We model a single "test cell" as a cube of side length L containing a Poisson-distributed number of dopants at mean density ρ (in atoms/cm^3).

$$\bar{N} = \rho L^3 \tag{A5}$$

is the mean number of dopants in the volume. For each Monte Carlo run we draw

$$N_\mathrm{neighbors} \sim \mathrm{Poisson}(\bar{N}) - 1, \tag{A6}$$

where the -1 accounts for the central atom at the origin

The $N_{\text{neighbors}}$ dopants are placed at random positions in the cube, uniformly distributed in

$$\mathbf{r}_i = (x_i, y_i, z_i) \sim \text{Uniform}\left[-\frac{L}{2}, \frac{L}{2}\right]^3, \quad i = 1, \ldots, N_{\text{neighbors}}, \tag{A7}$$

with the central atom fixed at $\mathbf{r} = 0$. Look at Fig. 12

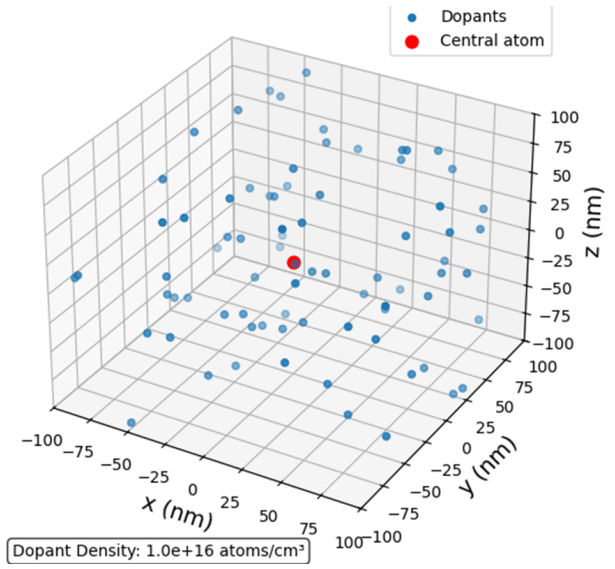


FIG. 12. Example Dopant Configuration set by **??**

3. Environmental Spin States and Orientation

Each dopant i is assigned a hyperfine manifold and Zeeman state based on simple "pumping-efficiency" parameters. We define:

- $F_{\text{pump}} = F_{\text{g}}$ and $m_{F,\text{pump}} = m_{F,\text{g}}$ as the desired pumped ground state;
- p_F as the probability of successfully pumping into the desired F_{pump};
- p_{m_F} as the probability of successfully pumping into the desired $m_{F,\text{pump}}$ within that manifold;
- p_{align} as the probability that the magnetic moment of a given dopant is aligned to the quantization axis $\hat{z}$.

For each dopant we then sample

$$F_i = \begin{cases} F_{\text{g}}, & \text{with probability } p_F, \\ F_{\text{e}}, & \text{with probability } 1 - p_F, \end{cases} \tag{A8}$$

$$m_{F,i} = \begin{cases} m_{F,\text{g}}, & \text{with probability } p_{m_F} \\ \text{uniform in } \{-F_{\text{g}}, \ldots, +F_{\text{g}}\}, & \text{otherwise.} \end{cases} \tag{A9}$$

The orientation of each dopant's magnetic moment is treated similarly. We take

$$\hat{\mathbf{m}}_i = \begin{cases} \hat{z}, & \text{with probability } p_{\text{align}}, \\ \text{random unit vector on } S^2, & \text{with probability } 1 - p_{\text{align}}. \end{cases} \tag{A10}$$

The magnitude of the magnetic moment is

$$m_i = \mu_B g_{F_i} m_{F,i}, \tag{A11}$$

so the full dipole moment vector is

$$\mathbf{m}_i = m_i \hat{\mathbf{m}}_i. \tag{A12}$$

4. Dipolar field and frequency shift

Each dopant is treated as a point magnetic dipole in free space. For a dipole $\mathbf{m}_i$ located at position $\mathbf{r}_i$ relative to the origin, the magnetic field at the origin is

$$\mathbf{B}_i(\mathbf{0}) = \frac{\mu_0}{4\pi r_i^3}\left[3\hat{\mathbf{r}}_i\,(\mathbf{m}_i \cdot \hat{\mathbf{r}}_i) - \mathbf{m}_i\right], \tag{A13}$$

where $\mathbf{r}_i = -\mathbf{r}_i^{(\text{dopant})}$ is the vector from the dipole to the origin, $r_i = |\mathbf{r}_i|$, and $\hat{\mathbf{r}}_i = \mathbf{r}_i / r_i$. We then sum over all dopants and project onto the quantization axis,

$$B_z = \hat{z} \cdot \sum_{i=1}^{N_{\text{neighbors}}} \mathbf{B}_i(\mathbf{0}). \tag{A14}$$

Given B_z, the frequency shift for that particular configuration is obtained from Eq. (A4),

$$\Delta\nu_k = \gamma_{\text{eff}} B_z^{(k)}, \tag{A15}$$

where $k = 1, \ldots, N_{\text{MC}}$ labels the Monte Carlo run.

5. Extracting linewidth and T_2^*

The inhomogeneous broadening is characterized by the distribution of frequency shifts $\Delta\nu_k$ over many random realizations of the spin environment. In the simulation we work in kHz units,

$$\Delta\nu_k^{(\text{kHz})} = 10^{-3}\,\Delta\nu_k^{(\text{Hz})}, \tag{A16}$$

and construct a histogram $H(\Delta\nu)$ from the Monte Carlo samples, Fig. 13 To extract an effective linewidth, we fit this histogram to a Lorentzian of the form

$$L(\nu) = A\,\frac{\Gamma/2}{(\nu - \nu_0)^2 + (\Gamma/2)^2}, \tag{A17}$$

where Γ is the full-width at half-maximum (FWHM) in kHz. The fitted FWHM Γ is then mapped to an inhomogeneous coherence time via

$$T_2^* = \frac{1}{\pi\Gamma}, \tag{A18}$$

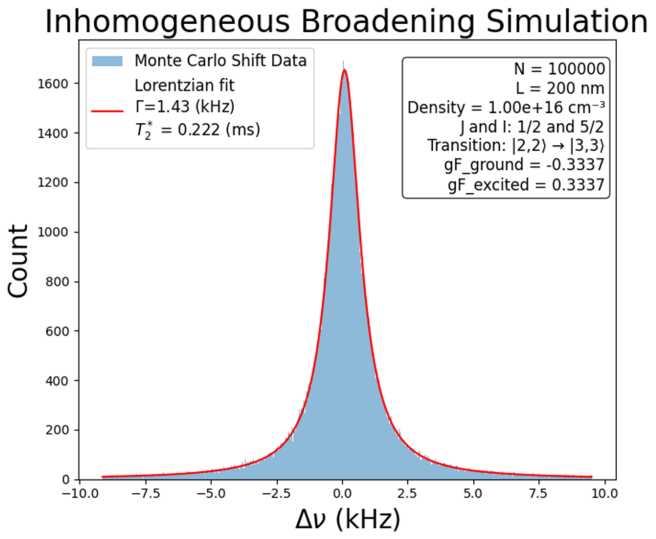


FIG. 13. Histogram, $H(\Delta\nu)$, from Monte Carlo samples of dipole-dipole interaction broadening

with Γ in kHz giving T_2^* in milliseconds. Equation (A18) is the standard relation between a Lorentzian FWHM and the corresponding exponential decay time.

Overall, this simulation: takes as inputs the density, box size, transition quantum numbers, and pumping/alignment efficiencies → generates N_{MC} independent configurations → computes the set of $\Delta\nu_k$ → performs the Lorentzian fit → extracts Γ and T_2^*.

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